Daniel Haehn, University of Massachusetts Boston haehn@cs.umb.edu haehn danielhaehn.com

I am a biomedical imaging and visualization researcher who investigates how computational methods can accelerate biological and medical research.

Education

2019 **PhD in Computer Science, Harvard University** Cambridge, MA
Analyzing Brain Connectivity and Computing Machine Perception
Advisor: Hanspeter Pfister
Committee: Steven Gortler, Finale Doshi-Velez, Scott Kuindersma, Jeff W. Lichtman

2010 **Diplom (MSc) in Medical Computer Science, University of Heidelberg** Germany
Signal- and Image Processing
Thesis: Coronary Artery Centerline Extraction
Advisors: Hartmut Dickhaus, Ron Kikinis

2007 **Vordiplom (BSc) in Medical Computer Science, University of Heidelberg** Germany
with Honors, rank #1 of class, all study fees waived

Experience

2019–present **University of Massachusetts Boston** Boston, MA
Assistant Professor of Computer Science (Tenure-track)
Director of the Machine Psychology research group
Director of the Artificial Intelligence (AI) Research Core Facility
Associate of the Harvard John A. Paulson School of Engineering and Applied Sciences

Summer 2017 **Apple, Inc.** Cupertino, CA
Research Intern in Data Science

Summer 2014 **Mental Canvas** New York City, NY
Research Intern in Computer Graphics

Experience (continued)

2011–2013 **Boston Children's Hospital** Boston, MA
Research Software Developer III, Fetal Neonatal Neuroimaging and Developmental Science Center
Advisors: Rudolph Pienaar, P. Ellen Grant

2010–2011 **University of Pennsylvania** Philadelphia, PA
Research Scholar, Section for Biomedical Image Analysis
Advisor: Kilian Pohl

2009 **German Cancer Research Center (DKFZ) and BioQuant Center** Heidelberg, Germany
Research Assistant, Biomedical Computer Vision and Experimental Radiology Research Groups
Advisors: Stefan Wörz, Hendrik von Tengg-Kobligk

2008–2009 **Brigham and Women's Hospital** Boston, MA
Fellow, Department of Radiology and the Surgical Planning Laboratory
Advisors: Ron Kikinis, Steve Pieper, Luca Antiga

Publications

From UMass Boston (* undergraduate student, ** graduate student; all peer-reviewed)

2025 Arianna Fernandez*, Sophia de Oliveira*, Chaya Spencer*, Kalysha Melendez Hernandez*, Samatrai Piam*, Daniel Haehn, and Amanda Potasznik. **CTRL + Empower: Lived Experiences of Black Women and Latinas in Computer Science at an Urban Public University**. *IEEE International Symposium on Ethics in Science, Technology and Engineering (ETHICS)*.

2025 Rami Huu Nguyen**, Kenichi Maeda*, Mahsa Geshvadi**, and Daniel Haehn. **Evaluating 'Graphical Perception' with Multimodal LLMs.** *IEEE Pacific Visualization (PacificVis)*.

2025 SangHyuk Kim**, Edward Gaibor*, Brian Matejek, and Daniel Haehn. **Melanoma Detection with Uncertainty Quantification**. *IEEE International Symposium on Biomedical Imaging (ISBI)*.

2025 Edward Gaibor*, Shruti Varade**, Rohini Deshmukh**, Tim Meyer**, Mahsa Geshvadi**, SangHyuk Kim**, Vidhya Sree Narayanappa**, and Daniel Haehn. **Boostlet.js: Medical Image Processing Plugins for the Web via JavaScript Injection**. *IEEE International Symposium on Biomedical Imaging (ISBI)*.

2025 Jiehyun Kim**, Kevin Wang, Yu Sakai, Youxiang Zhu, Andrew C. Hu, Huy Q. Phi, Nathan Arnett, Grace J. Wang, Brett L. Cicchiara, Jae W. Song, and Daniel Haehn. **Automatic Segmentation of Calcified Plaque in Carotid Arteries**. *IEEE International Symposium on Biomedical Imaging (ISBI)*.

2024 Youxiang Zhu**, Nana Lin**, Kiran Sandilya Balivada**, Daniel Haehn, and Xiaohui Liang. **Adversarial Text Generation using Large Language Models for Dementia Detection**. *Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

2024 Tim Meyer**, Gabi Dreo Rodosek, and Daniel Haehn. **WebGL-based Image Processing through JavaScript Injection**. *ACM Conference on 3D Web Technology*.

2024 Zhenxing Cui, Lu Chen, Yunhai Wang, Daniel Haehn, Yong Wang, and Hanspeter Pfister. **Generalization of CNNs on Relational Reasoning With Bar Charts.** *IEEE Transactions on Visualization and Computer Graphics*.

2024 Amanda R. Potasznik, Funda Durupinar, and Daniel Haehn. **'Just let me': Experiences of Historically Marginalized Students in the CS Department**. *IEEE Black Issues in Comp. Education (BICE)*.

2024 Loraine Franke**, Tae Young Park, Jie Luo, Yogesh Rathi, Steve Pieper, Lipeng Ning, and Daniel Haehn. **SlicerTMS: Real-Time Visualization of Transcranial Magnetic Stimulation for Mental Health Treatment**. *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*.

2024 Charlotte Kyeremah**, Matthew Weiss**, Dila Kandel**, Daniel Haehn, and Chandra Yelleswarapu. **Single-beam digital holographic reconstruction: a phase-support enhanced complex wavefront on phase-only function for twin-image elimination.** *Journal of Biomedical Optics*.

2024 Loraine Franke**, Daniel Karl I. Weidele, Nima Dehmamy, Lipeng Ning, and Daniel Haehn. **AutoRL X: Automated Reinforcement Learning on the Web.** *ACM Transactions Interactive Intelligent Systems*.

2024 Akshata Tiwari*, Brian Matejek, and Daniel Haehn. **Non-invasive Stress Monitoring from Video**. *IEEE International Symposium on Biomedical Imaging (ISBI)*.

2024 Tae Young Park, Loraine Franke**, Steve Pieper, Daniel Haehn, and Lipeng Ning. **A review of algorithms and software for real-time electric field modeling techniques for transcranial magnetic stimulation.** *Biomedical Engineering Letters*.

2023 Navaneethakrishna Makaram, Sarvagya Gupta**, Matthew Pesce, Jeffrey Bolton, Scellig Stone, Daniel Haehn, Marc Pomplun, Christos Papadelis, Phillip Pearl, Alexander Rotenberg, Patricia Ellen Grant, and Eleonora Tamilia. **Deep Learning-Based Visual Complexity Analysis of Electroencephalography Time-Frequency Images: Can It Localize the Epileptogenic Zone in the Brain?**. *MDPI Algorithms*.

2023 Kristin Qi**, Jiali Cheng, and Daniel Haehn. **Lesion Search with Self-supervised Learning.** *International Conference on Learning Representations (ICLR)*.

2023 Ryan Zurrin*, Neha Goyal**, Pablo Bendiksen**, Muskaan Manocha**, Dan Simovici, Nurit Haspel, Marc Pomplun, and Daniel Haehn. **Outlier Detection for Mammograms**. *International Conference on Medical Imaging with Deep Learning (MIDL)*.

2023 Ramin Dehghanpoor**, Fatemeh Afrasiabi**, Charles Fogel*, Tung Dao*, Suman Gautam*, Aanab Nehela*, Ahmad Nehela*, Daniel Haehn, and Nurit Haspel. **Classifying Protein Families with Learned Compressed Representations**. *International Conference on Bioinformatics (Best Paper Award at BICOB)*.

2023 Daniel Karl I. Weidele, Shazia Afzal, Abel N. Valente, Cole Makuch, Owen Cornec, Long Vu, Dharmashankar Subramanian, Werner Geyer, Rahul Nair, Inge Vejsbjerg, Radu Marinescu, Paulito Palmes, Elizabeth M. Daly, Loraine Franke**, and Daniel Haehn. **AutoDOViz: Human-Centered Automation for Decision Optimization**. *ACM International Conference on Intelligent User Interfaces (IUI)*.

2022 Neha Goyal**, Yahiya Hussain*, Gianna G. Yang*, and Daniel Haehn. **Real-Time Alignment for Connectomics**. *Springer LNCS: Biomedical Image Registration (WBIR)*.

2022 Francois Rheault, Valérie Hayot-Sasson, Robert E. Smith, Christopher Rorden, Jacques-Donald Tournier, Eleftherios Garyfallidis, Fang-Cheng Yeh, Christopher J. Markiewicz, Matthew Brett, Ben Jeurissen, Paul A. Taylor, D. Baran Aydogan, Derek A. Pisner, Serge Koudoro, Soichi Hayashi, Daniel Haehn, Steve Pieper, Daniel Bullock, Emanuele Olivetti, Jean-Christophe Houde, Marc-Alexandre Côté, Flavio Dell'Acqua, Alexander Leemans, Maxime Descoteaux, Bennett Landman, Franco Pestilli, and Ariel Rokem. **TRX: A community-oriented tractography file format**. *OHBM 2022-Human Brain Mapping (Oral)*.

2022 Jay Burkhardt*, Aaryaman Sharma*, Jack Tan*, Loraine Franke**, Jahnavi Leburu**, Jay Jeschke, Sasha Devore, Daniel Friedman, Jingyun Chen, and Daniel Haehn. **N-Tools-Browser: Web-Based Visualization of Electrocorticography Data for Epilepsy Surgery**. *Frontiers in Bioinformatics*.

2022 Katharina Paulick, Simon Seidel, Christoph Lange, Annina Kemmer, Mariano Nicolas Cruz-Bournazou, André Baier, and Daniel Haehn. **Promoting Sustainability through Next-Generation Biologics Drug Development**. *MDPI Sustainability*.

2022 Nalini M. Singh, Jordan B. Harrod, Sandya Subramanian, Mitchell Robinson, Ken Chang, Suheyla Cetin-Karayumak, Adrian Vasile Dalca, Simon Eickhoff, Michael Fox, Loraine Franke**, Polina Golland, Daniel Haehn, Juan Eugenio Iglesias, Lauren J. O'Donnell, Yangming Ou, Yogesh Rathi, Shan H. Siddiqi, Haoqi Sun, M. Brandon Westover, Susan Whitfield-Gabrieli, and Randy L. Gollub. **How Machine Learning is Powering Neuroimaging to Improve Brain Health**. *Neuroinformatics*.

2021 Bella Baidak*, Yahiya Hussain*, Emma Kelminson*, Thouis R. Jones, Loraine Franke**, and Daniel Haehn. **CellProfiler Analyst Web (CPAW) - Exploration, analysis, and classification of biological images on the web**. *IEEE Visualization Short Paper (IEEE VIS)*.

2021 Loraine Franke**, Daniel Karl I Weidele, Fan Zhang, Suheyla Cetin-Karayumak, Steve Pieper, Lauren J O'Donnell, Yogesh Rathi, and Daniel Haehn. **FiberStars: Visual Comparison of Diffusion Tractography Data between Multiple Subjects**. *IEEE Pacific Visualization (PacificVis)*.

2020 Loraine Franke** and Daniel Haehn. **Modern Scientific Visualizations on the Web**. *MDPI Informatics*.

2020 Daniel Haehn, Loraine Franke**, Fan Zhang, Suheyla Cetin Karayumak, Steve Pieper, Lauren O'Donnell, and Yogesh Rathi. **TRAKO: Efficient Transmission of Tractography Data for Visualization**. *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*.

2020 Vincent Casser, Kai Kang, Hanspeter Pfister, and Daniel Haehn. **Fast Mitochondria Detection for Connectomics**. *International Conference on Medical Imaging with Deep Learning (Spotlight Award at MIDL)*.

2020 Zudi Lin, Donglai Wei, Won-Dong Jang, Siyan Zhou, Xupeng Chen, Xueying Wang, Richard L. Schalek, Daniel R. Berger, Brian Matejek, Lee D. Kamentsky, Adi Peleg, Daniel Haehn, Thouis R. Jones, Toufiq Parag, Jeff W. Lichtman, and Hanspeter Pfister. **Two-Stream Active Query Suggestion for Large-Scale Object Detection in Connectomics**. *European Conference on Computer Vision (ECCV)*.

2020 Fritz Lekschas, Brant Peterson, Daniel Haehn, Eric Ma, Nils Gehlenborg, and Hanspeter Pfister. **Peax: Interactive Visual Pattern Search in Sequential Data Using Unsupervised Deep Representation Learning.** *Computer Graphics Forum (Best Paper Award at EuroVis)*.

Prior to UMass Boston

2019 Brian Matejek, Daniel Haehn, Haidong Zhu, Donglai Wei, Toufiq Parag, and Hanspeter Pfister. **Biologically-Constrained Graphs for Global Connectomics Reconstruction**. *IEEE Computer Vision and Pattern Recognition (CVPR)*.

2018 Daniel Haehn, James Tompkin, and Hanspeter Pfister. **Evaluating 'Graphical Perception' with CNNs.** *IEEE Transactions on Visualization and Computer Graphics (IEEE VIS)*.

2018 Daniel Haehn, Verena Kaynig, James Tompkin, Jeff W. Lichtman, and Hanspeter Pfister. **Guided Proofreading of Automatic Segmentations for Connectomics**. *IEEE Computer Vision and Pattern Recognition (CVPR)*.

2017 Daniel Haehn, John Hoffer, Brian Matejek, Adi Suissa-Peleg, Ali K. Al-Awami, Lee Kamentsky, Felix Gonda, Eagon Meng, William Zhang, Richard Schalek, Alyssa Wilson, Toufiq Parag, Johanna Beyer, Verena Kaynig, Thouis R. Jones, James Tompkin, Markus Hadwiger, Jeff W. Lichtman, and Hanspeter Pfister. **Scalable Interactive Visualization for Connectomics.** *MDPI Informatics*.

2017 Brian Matejek, Daniel Haehn, Fritz Lekschas, Michael Mitzenmacher, and Hanspeter Pfister. **Compresso: Efficient Compression of Segmentation Data For Connectomics**. *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*.

2017 Felix Gonda, Verena Kaynig, Thouis R. Jones, Daniel Haehn, Jeff W. Lichtman, Toufiq Parag, and Hanspeter Pfister. **ICON: An Interactive Approach to train Deep Neural Networks for Segmentation of Neuronal Structures**. *IEEE International Symposium on Biomedical Imaging (ISBI)*.

2017 Rudolph Pienaar, Ata Turk, Jorge Bernal-Rusiel, Nicolas Rannou, Daniel Haehn, P. Ellen Grant, and Orran Krieger. **CHIPS--A Service for Collecting, Organizing, Processing, and Sharing Medical Image Data in the Cloud**. *VLDB Workshop on Data Management and Analytics for Medicine and Healthcare*.

2016 Adi Suissa-Peleg, Daniel Haehn, Seymour Knowles-Barley, Verena Kaynig, Thouis R. Jones, Alyssa Wilson, Richard Schalek, Jeff W. Lichtman, and Hanspeter Pfister. **Automatic Neural Reconstruction from Petavoxel of Electron Microscopy Data.** *Microscopy and Microanalysis*.

2016 Ali K. Al-Awami, Johanna Beyer, Daniel Haehn, Narayanan Kasthuri, Jeff W. Lichtman, Hanspeter Pfister, and Markus Hadwiger. **NeuroBlocks--Visual Tracking of Segmentation and Proofreading for Large Connectomics Projects.** *IEEE Transactions on Visualization and Computer Graphics (IEEE VIS)*.

2016 Richard Schalek, Dong Lee, Narayanan Kasthuri, Adi Peleg, Thouis R. Jones, Verena Kaynig, Daniel Haehn, Hanspeter Pfister, David Cox, and Jeff W. Lichtman. **Imaging a 1 mm^3 Volume of Rat Cortex using a Multi-Beam SEM.** *Microscopy and Microanalysis*.

2015 Kiho Im, Banu Ahtam, Daniel Haehn, Jurriaan M. Peters, Simon K. Warfield, Mustafa Sahin, and P. Ellen Grant. **Altered Structural Brain Networks in Tuberous Sclerosis Complex.** *Cerebral Cortex*.

2015 Rudolph Pienaar, Nicolas Rannou, Jorge Bernal, Daniel Haehn, and P. Ellen Grant. **ChRIS--A web-based Neuroimaging and Informatics System for Collecting, Organizing, Processing, Visualizing and Sharing of Medical Data.** *IEEE Engineering in Medicine and Biology Society (EMBC)*.

2014 Daniel Haehn, Seymour Knowles-Barley, Mike Roberts, Johanna Beyer, Narayanan Kasthuri, Jeff W. Lichtman, and Hanspeter Pfister. **Design and Evaluation of Interactive Proofreading Tools for Connectomics.** *IEEE Transactions on Visualization and Computer Graphics (IEEE VIS)*.

2013 Daniel Haehn, Nicolas Rannou, P. Ellen Grant, and Rudolph Pienaar. **Slice:Drop -- Collaborative Medical Imaging in the Browser.** *ACM SIGGRAPH Computer Animation Festival*.

2012 Daniel Haehn, Nicolas Rannou, Banu Ahtam, P. Ellen Grant, and Rudolph Pienaar. **Neuroimaging in the Browser using the X Toolkit.** *Frontiers in Neuroinformatics (Spotlight Award at INCF Neuroinformatics)*.

2012 Myong-sun Choe, Silvia Ortiz-Mantilla, Nikos Makris, Matt Gregas, Janine Bacic, Daniel Haehn, David Kennedy, Rudolph Pienaar, Verne S. Caviness Jr, April A. Benasich, and P. Ellen Grant. **Regional Infant Brain Development: an MRI-based Morphometric Analysis in 3 to 13 month olds.** *Cerebral Cortex*.

2012 Arno Klein, Forrest S. Bao, Yrjö Häme, Eliezer Stavsky, Joachim Giard, Daniel Haehn, Nolan Nichols, and Satrajit S. Ghosh. **Mindboggle: Automated Human Brain MRI Feature Extraction, Labeling, Morphometry, and Online Visualization.** *Frontiers in Neuroinformatics*.

2012 Arno Klein, Nolan Nichols, and Daniel Haehn. **Mindboggle 2 interface: Online Visualization of Extracted Brain Features with XTK.** *Frontiers in Neuroinformatics*.

Grants

Funded (* Principal Investigator)

2025–2027 *Sloan Foundation Grant: **Expanding Culture Change across STEM Disciplines at the University of Massachusetts Public Education System: A Partnership Between UMass Boston (UMB) and Amherst (UMA)**, (Co-PIs UMB: Kimberly Hamad Schifferli, UMA: Neena Thota, Shannon Roberts), $500,000.00

2023–2025 National Institutes of Health, U54 Supplement: **iTCGA---Integrated Training in Computational Genomics and Data Sciences**, Co-PI (PI Adan Colon-Carmona, PI Jill Mascoska, Co-PI Brooke Moyers, Co-PI Kourosh Zarringhalam, Co-PI Christina Kelliher; UMB), $581,237.00

2022–2024 Sloan Foundation Grant: **Culture Change in Computer Science and Engineering at the University of Massachusetts Public University System: A Partnership Between UMass Boston and Amherst**, UMB Site-PI (Co-PI Hamad Schifferli, UMA: PI Dasgupta, Co-PI Thota, Co-PI Roberts), $499,972.00 (UMB share $238,948.00)

2021–2023 National Institutes of Health, R21: **Real-time visualization and precision targeting in transcranial magnetic stimulation**, Co-PI (PI Lipeng Ning, Harvard Medical School), $493,011.00 (UMB share $156,663.00)

2021–2022 UMass Boston, Proposal Development Grant: **Towards Developing Deep Learning Approaches for Protein-Protein Interaction Detection**, Co-PI (PI Nurit Haspel, UMB), $20,000.00

2020–2023 *Massachusetts Life Sciences Center, Bits to Bytes: **The Oregon-Massachusetts Mammography Database (OMAMA-DB)**, (Co-PIs Haspel, Tonyushkin, Pomplun, Simovici), $749,834.00

2020 Federal Ministry of Education and Research Germany: International Future Labs for Artificial Intelligence in collaboration with the KIWI Biolab at the Technical University Berlin (covering 18 months exchange visits of a PostDoc and a Ph.D. student), (PI Cruz-Bournazou, TU Berlin)

2019 *NVidia Accelerated Data Science GPU Grant, 1x Titan V100 GPU

Presentations

* invited presentation

2025 *Speaker at **Visualizing Biological Data (VIZBI)**: *Masterclass on Web Graphics*, University of Cambridge, UK

2025 *Panelist for **Employment Opportunities in the Age of Cyber Technology Proliferation and AI** at Brandeis University

2025 ***AI Core Capabilities at the Venture Development Center** at UMass Boston

Presentations (continued)

2025	**National Research Mentoring Network** Instructor for Graduate Students at UMass Boston
2024	*__High-Performance Computing and AI Core Presentation__ for the Healey Library at UMass Boston
2024	Speaker at the **Sloan Faculty Learning Community**: *Difficult Conversations*
2024	*__Center for Innovative Teaching Presentation of Inclusive Pedagogies__ at UMass Boston
2024	**National Research Mentoring Network** Instructor for Faculty at UMass Boston
2024	*Panelist at **TEACH**: *Creative Assignments in the age of AI*
2023	Speaker at the **Sloan Faculty Learning Community**: *Difficult Conversations*
2023	*Speaker at **MIT Brainhack**: *TRAKO and beyond*
2023	*Speaker at the **Hack NiiVue.js!, University of South Carolina**: *BOOSTLET.js*
2022	*Speaker at the **High Performance Computing Day**: *Processing of Massive Biological Datasets at UMB*
2022	*Speaker at **Visualizing Biological Data (VIZBI)**: *Masterclass on Scientific Visualization*
2021	*Presenter at the **UMass Summit for AI, Data Science, and Robotics**
2020	*Presenter at the **Creative Commons Global Summit**: *The 7 Levels of Open Science*
2020	Presenter at the **National Alliance for Medical Image Computing Project Week**: *Integrating TRAKO with 3D Slicer*
2020	*Speaker at the **Fetal Neonatal Developmental Science Center, Boston Children's Hospital**: *Scientific Visualization at Scale!*
2020	Paper presentation at **International Conference on Medical Image Computing and Computer Assisted Intervention**: *TRAKO: Efficient Transmission of Tractography Data for Visualization*
2020	*Speaker at the **Lymph Node Quantification Project, Harvard Medical School**: *Machine-Guided Annotation Methods*
2020	Paper presentation at **Medical Imaging with Deep Learning (MIDL)**: *Fast Mitochondria Detection for Connectomics*
2020	*Presentation at the **UMass Boston-Dana Farber/Harvard Cancer Center initiative**: *Guided Tumor Detection and Annotation Methods for Cancer Imaging*
2020	*Speaker at the **Massachusetts Life Sciences Center**: *The Oregon-Massachusetts Mammography Database*
2020	*Researcher at **Shonan Meeting No. 167 in Japan**: *Formalizing Biological and Medical Visualization*
2019	*Speaker at **Sarah Frisken's Lab, Harvard Medical School**: *Brain Connectivity, Machine Perception, and Computer Graphics - all at different scales!*
2019	*Speaker at **Suffolk University**: *Brain Connectivity and Machine Perception*
2019	*Speaker at the **MIT McGovern Institute**: *The Performance Gap between the Brain and AI*
2018	Paper presentation at **IEEE Visualization**: *Evaluating 'Graphical Perception' with CNNs*
2018	**Harvard Visual Computing Group** meeting presentation: *The 7 Levels of Open Science*
2018	*Speaker at **Brown University, Department of Computer Science**: *Analyzing Brain Connectivity and Computing Machine Perception*

Presentations (continued)

2018	*Speaker at **IBM Research** (AI Systems Day): *Evaluating 'Graphical Perception' with CNNs*
2017	**Harvard Visual Computing Group** meeting presentation: *Guided Proofreading of Automatic Segmentations for Connectomics*
2016	*Speaker at the **IEEE Visualization Doctoral Colloquium**: *Proofreading for Connectomics*
2015	**Harvard Lichtman Lab** meeting presentation: *Interactive Proofreading Tools for Connectomics*
2014	Paper presentation at **IEEE Visualization**: *Design and Evaluation of Interactive Proofreading Tools for Connectomics*
2014	**Harvard Visual Computing Group** meeting presentation: *Proofreading Tools for Connectomics*
2014	*Speaker at the **MIT Computer Graphics Group**: *Web-based Visualization of Scientific Data*
2014	**Harvard Visual Computing Group** meeting presentation: *Interactive Proofreading with Dojo*
2014	**Harvard Lichtman Lab** meeting presentation: *Web-based Visualization and Proofreading for Connectomics*
2013	**Harvard Visual Computing Group** meeting presentation: *Web-based Scientific Visualization*
2013	*Speaker at **Visualizing Biological Data (VIZBI)**: *Physiology & Function*
2012	Spotlight presentation at **INCF Neuroinfomatics**: *Neuroimaging in the Browser using the X Toolkit*
2012	*Speaker at **WebGL Camp Orlando**: *WebGL for Baby Brains*

Awards

2024	Wasabi Fenway Bowl, Educator Shoutout at Fenway Park
2024	Excellence in Undergraduate Education Award in the College of Science and Mathematics at UMass Boston
2024	Center for Innovative Teaching (CIT) Award for Inclusive Pedagogies at UMass Boston
2023	Early Career Research Excellence Award in the College of Science and Mathematics at UMass Boston
2023	Best Paper Award at BICOB for Classifying Protein Families
2022	Selected for Oral Presentation at ISMRM Neuromodulation for TMS Visualization
2022	Selected for Oral Presentation at the Organization for Human Brain Mapping for TRX
2020	Best Paper Award at EuroVis for Peax
2020	Spotlight Award at MIDL: Fast Mitochondria Detection for Connectomics
2020	AI Scientist of the Future for the KIWI Biolab at the Technical University Berlin, Germany
2015–2019	Winkler Scholarship
2013–2019	Harvard University Fellowship
2013	Real-Time Live! presentation of Slice:Drop at SIGGRAPH
2012	INCF Neuroinformatics Spotlight Award for XTK
2012	Mozilla Hacks WebGL Dev Derby Runner-up for Slice:Drop
2012	Visualizing.org VisWeek Challenge Winner with Slice:Drop

Awards (continued)

2010	1st Prize for End User Tutorial at the National Alliance of Medical Image Computing (NA-MIC)
2008–2009	Karl Steinbuch Foundation Scholarship
2007–2009	Thomas Gessmann Foundation Scholarship

Teaching

At UMass Boston (main instructor unless indicated, * re-designed course, ** new course)

2024	CS460 Graphics (40 students, Instructor rating: 4.93/5, Course rating: 4.9/5)
2024	**CS480/CS697 Special Topics: Visualizing.Boston (20 students, Instructor rating: 5/5, Course rating: 5/5)
2024	CS666 Biomedical Signal and Image Processing (19 students, Instructor rating: 4.68/5, Course rating: 4.63/5)
2023	CS460 Graphics (48 students, Instructor rating: 4.89/5, Course rating: 4.85/5)
2023	**CS666 Biomedical Signal and Image Processing (57 students, Instructor rating: 4.92/5, Course rating: 4.82/5)
2023	CS410 Introduction to Software Engineering (60 students, Instructor rating: 4.67/5, Course rating: 4.44/5)
2023	Guest Lecturer for CS615 User Interface Design
2023	Guest Lecturer at the Integrative Biosciences & Computational Sciences Graduate Program Seminar
2022	CS460 Graphics (48 students, Instructor rating: 4.76/5, Course rating: 4.79/5)
2022	Guest Lecturer for CS615 User Interface Design
2022	CS480/CS697 Special Topics: Biomedical Signal and Image Processing (27 students, Instructor rating: 5/5, Course rating: 4.93/5)
2022	CS410 Introduction to Software Engineering (62 students, Instructor rating: 4.23/5, Course rating: 4.41/5)
2022	Guest Lecturer for BIOL693 Seminar in Neurobiology
2021	Guest Lecturer for CS615 User Interface Design
2021	CS460 Graphics (20 students, Instructor rating: 5/5, Course rating: 4.94/5)
2021	**CS480/CS697 Special Topics: Biomedical Signal and Image Processing (27 students, Instructor rating: 4.8/5, Course rating: 4.75/5)
2021	CS410 Introduction to Software Engineering (47 students, Instructor rating: 4.59/5, Course rating: 4.45/5)
2020	CS460 Graphics (28 students, Instructor rating: 4.9/5, Course rating: 4.9/5)
2020	*CS410 Introduction to Software Engineering (27 students, Instructor rating: 4.87/5, Course rating: 4.73/5)
2019	**CS460 Graphics (24 students, Instructor rating: 4.81/5, Course rating: 4.57/5)
2019	Guest Lecturer for two lectures of the CS187 Science Gateway Seminar

Outside UMass Boston

2021	Guest Lecturer for CSCI2254 Web Application Development at Boston College
2020	Guest Lecturer for CSCI2254 Web Application Development at Boston College
2019	Guest Lecturer for the CMPSC131 Computer Science course at Suffolk University
2018–2019	TEALS Volunteer for AP Computer Science at Cambridge Rindge and Latin School
2016	Technical Assistant for the Deep Learning mini-course at the Harvard IACS Compute Fest
2015	Teaching Fellow for the Harvard CS171 Visualization course
2008	Workshop for Advanced Microcontroller Programming, University of Bratislava, Slovakia
2008	Workshop for Microcontroller Programming at the University of Tbilisi, Georgia (Europe)
2004–2008	Teaching Assistant for the Microcontrollers in EXperiment and LEarning (MEXLE) educational platform, Heilbronn University, Germany

Mentoring

Graduate Students (PhD)

2024–present	Rami Huu Nguyen, Computational Sciences, University of Massachusetts Boston
2023–present	SangHyuk Kim, Computer Science, University of Massachusetts Boston
2022–2024	Mahsa Geshvadi, Computer Science, University of Massachusetts Boston
2022–present	Jenna (JieHyun) Kim, Computer Science, University of Massachusetts Boston
2020–2023	Kristin (Yanan) Qi, Computational Sciences, University of Massachusetts Boston
2021	Hayoun Oh, Computer Science, Harvard University (co-mentored)
2020–2021	Aswin Vasudevan, Computer Science, University of Massachusetts Boston
2019–2024	Loraine Franke, Computer Science, University of Massachusetts Boston
2019–2021	Jesse Freeman, Computer Science, University of Massachusetts Boston

Graduate Students (MS)

2023–2025	Shruti Shailendra Varade, Computer Science, University of Massachusetts Boston
2023–2025	Vidhya Sree Narayanappa, Computer Science, University of Massachusetts Boston
2022–2024	Dhruv Shah, Computer Science, University of Massachusetts Boston
2022–2023	Kunal Jain, Computer Science, University of Massachusetts Boston
2022–2025	Kiran Sandilya, Computer Science, University of Massachusetts Boston
2022	Benni Bendiksen, Computer Science, University of Massachusetts Boston
2021–2023	Neha Goyal, Computer Science, University of Massachusetts Boston
2020	Jiali Cheng, Computer Science, Northeastern University

Mentoring (continued)

2020	Gianna Yang, Computer Science, University of Massachusetts Boston
2020	Barkha Java, Computer Science, University of Massachusetts Boston
2019	Manish Mourya, Computer Science, University of Massachusetts Boston
2018–2020	Vincent Casser, Computer Science, Harvard University
2010-2011	Suares Tamekue, Intern at Brigham and Women's Hospital (co-mentored)

Undergraduate and Pre-College Students

2024–present	Blake Moody, Computer Science, University of Massachusetts Boston
2024–present	Daniel Gross, Computer Science, University of Massachusetts Boston
2024–present	Kenichi Maeda, Computer Science, University of Massachusetts Boston
2023–present	Edward Gaibor, Computer Science, University of Massachusetts Boston
2022–2024	Ryan Zurrin, Computer Science, University of Massachusetts Boston
2023	Amanda Kwong, Conputer Science and Honors Thesis, University of Massachusetts Boston
2022–2023	Josh Kotler, Computer Science, University of Massachusetts Boston
2022–present	Akshata Tiwari, Pre-college student at Aliso Niguel High School
2022	Kendrick Kheav, Biochemistry, University of Massachusetts Boston
2022	Nikol Vladinska, Computer Science and Honors Thesis, University of Massachusetts Boston
2021–2022	Jay Burkhardt, Computer Science, University of Massachusetts Boston
2021	Isabelle Lara, Biology, University of Massachusetts Boston
2021	Patricia Somera, Biology, University of Massachusetts Boston
2021	Bella Baidak, Computer Science, University of Massachusetts Boston
2019–2021	Yahiya Hussain, Computer Science, University of Massachusetts Boston
2019–2021	Nandinii Yeleswarapu, Computer Science and Honors Thesis, University of Massachusetts Boston
2019–2020	Safwa Ali, Engineering, University of Massachusetts Boston
2019–2020	Huda Irshad, Engineering, University of Massachusetts Boston
2018–2020	Ian Svetkey, Pre-college student at Harvard University
2016	Eagon Meng, Computer Science, Harvard University
2016	Omar Shaikh, (Remote-) Intern at Harvard University
2015–2017	John Hoffer, Computer Science, Harvard University
2015	William Zhang, Pre-college student at Harvard University
2013	Jay Andrew Robinson, Intern at Boston Children's Hospital (co-mentored)
2013	Emily Seibring, Intern at Boston Children's Hospital (co-mentored)

Service and Outreach

Departmental Level

2023–present	Chair of the Student Activities Committee
2022–present	Member of the Broadening Participation in Computing Task Force
2022–present	Faculty Advisor for the Computer Science Club
2020–present	Organizer of Events and Maintainer of the Discord Server for CS+IT Students
2020	Member of the Paul English Scholarship Committee
2019–present	Member of the Outreach and Publicity Committee
2019–present	Member of the Student Recruitment Committee
2019–2020	Organizer of bi-weekly social events for IT and CS students

College and University Level

2024–present	Faculty Advisor for the UMass Boston Chapter of the National Society of Black Engineers (NSBE)
2024	Member of the Research Computing Hiring Committee
2024	Chair of the Poster Judging Committee for the CSM Student Success Showcase
2024	Member of the Faculty Search Committee for Physics-based Machine Learning
2024	Member of the CSM Student Success Showcase Organization Committee
2023	Judge for the CSM Student Success Showcase Poster Awards
2023	Member of the Research Computing Hiring Committee
2022	Invited Researcher for the UMass President's Office NSF Type 2 Engine proposal for Equitable Health
2022	Member of the Research Computing Hiring Committee
2022	Member of the Data Science Faculty Search Committee
2022	Member of the Joint Discipline & Grievance Committee
2021	Panelist for the Brain Trusts in AI/Robotics/Data Science Initiative from the UMass President's Office
2020–present	Member of the Research Computing Advisory Committee
2020–2021	STEM Educational Excellence (STEM-EdX) Fellow
2020	Member of the Data Science Faculty Search Committee

Professional Activities

2025	Reviewer for the Health Equity Accelerator at the Massachusetts Life Sciences
2024	White House Summit on STEMM Equity and Excellence, Leveraging Diverse Minds in R&D Discussion
2024	Host and Organizer of the first Boston Brainhack event at UMass Boston
2023	Trained Instructor for the National Research Mentoring Network (NRMN)

Service and Outreach (continued)

2023	Reviewer for the MICCAI Grand Challenge for Aorta Segmentation
2023	Public Interest Technology (PIT) New England Member
2022	National Science Foundation Reviewer and Panelist for SBIR Grants
2022	Program Committee member at the IEEE Visualization conference
2022	Topic Editor: Open Source for Open Science, Frontiers in Neuroinformatics
2021	Program Committee member at the IEEE Visualization conference
2021	Faculty Mentor at the MGH Neuroimaging 2021 Virtual Symposium
2021	Organizer of the Chart Question Answering Workshop at CVPR 2021 in collaboration with Harvard, Columbia, Northwestern, and UMass Amherst
2021	National Science Foundation Reviewer and Panelist for SBIR Grants
2020	Program Committee member for short papers at the IEEE Visualization conference
2018–present	Reviewer for *Manning Publications*
2016–present	Reviewer for Frontiers in Neuroinformatics, ISMRM, Neuroinformatics, Frontiers in Neural Circuits, ACM SIGCHI, IEEE CVPR, ECCV, IEEE Visualization / Transactions on Visualization and Computer Graphics, IEEE Access, MDPI Applied Sciences, Nature Communications Biology, Scientific Reports, Transactions on Pattern Analysis and Machine Intelligence, Nature, Computer & Graphics
2013	Technical Reviewer for *Matsuda and Lea: WebGL Programming Guide, Addison-Wesley*

Community Service

2025–present	Public Interest Technology New England (PIT-NE) Leadership Committee
2024–present	Public Interest Technology New England (PIT-NE) Ambassador for UMass Boston
2023	Volunteer for the Petey Greene Program to tutor incarcerated people
2022	Member of the Principal Search Committee for the Putnam Ave Upper School in Cambridge
2019–2020	Advisor for the AP Data Science Curriculum in Cambridge Public Schools
2018–2019	Head Coach for Cambridge Youth Soccer
2018	Volunteer+Presentation Facilitator at the Cambridge 8th Grade Science & Engineering Showcase
2007–2010	President of the Student Computer Club at Heilbronn University, StuWoNet e.V.
2007–2009	Voluntary Project Lead of RANDI2, a randomization software for clinical trials at the German Cancer Research Center (DKFZ), coordinating 15+ developers
1997–1999	Vice-President of The German Computer Freaks, a National Cyber Security Club

My Erdős Number is 3.